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FOVEA

Detection of Autism Indicators in Children

A Jonior Project Report

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Abstract

Autism spectrum disorder is a neurodevelopmental condition characterized by a wide range of behavioral and movement-related patterns. Early observation of such behaviors plays a critical role in supporting timely intervention. This project proposes a computer vision–based system for detecting autism-related behavioral indicators in children using the YOLO v11 object detection framework.

The system analyzes visual data extracted from videos to identify movement-based behaviors associated with autism. Due to the limited availability of publicly accessible data, a custom dataset was manually annotated in compliance with ethical and legal considerations. Data augmentation techniques were applied to enhance model generalization. The model was trained and evaluated using standard object detection metrics, with particular emphasis on recall to minimize missed detections.

Experimental results demonstrate that the proposed system achieved a detection performance of approximately 86%, indicating its ability to effectively identify autism-related behaviors despite dataset limitations. While the system is not intended for diagnostic use, it shows strong potential as a supportive tool for behavioral analysis and early screening. The project highlights both the feasibility and challenges of applying deep learning–based object detection to autism-related behavior analysis.

Abstract

يُعد اضطراب طيف التوحد حالة نمائية عصبية تتميز بمجموعة واسعة من الأنماط السلوكية والحركية. وتلعب الملاحظة المبكرة لهذه السلوكيات دورًا حاسمًا في دعم التدخل المبكر في الوقت المناسب. يقترح هذا المشروع نظامًا قائمًا على الرؤية الحاسوبية لاكتشاف المؤشرات السلوكية المرتبطة بالتوحد لدى الأطفال باستخدام إطار عمل YOLO v11 لاكتشاف الأجسام.

يقوم النظام بتحليل البيانات المرئية المستخرجة من مقاطع الفيديو لتحديد السلوكيات الحركية المرتبطة بالتوحد. ونظرًا لمحدودية توفر البيانات المتاحة للعامة، تم إعداد مجموعة بيانات مخصصة من خلال إجراء عملية ترميز (Annotation) يدوية مع الالتزام بالاعتبارات الأخلاقية والقانونية. كما تم تطبيق تقنيات تعزيز البيانات لتحسين قدرة النموذج على التعميم. تم تدريب النموذج وتقييمه باستخدام معايير اكتشاف الأجسام القياسية، مع التركيز بشكل خاص على مقياس الاستدعاء (Recall) بهدف تقليل حالات الفشل في الاكتشاف.

أظهرت النتائج التجريبية أن النظام المقترح حقق أداء كشف يقارب 86%، مما يشير إلى قدرته على التعرف بفعالية على السلوكيات المرتبطة بالتوحد رغم محدودية مجموعة البيانات. وعلى الرغم من أن النظام غير مخصص للاستخدام التشخيصي، إلا أنه يُظهر إمكانات قوية كأداة داعمة لتحليل السلوك والكشف المبكر. كما يسلط المشروع الضوء على كلٍّ من جدوى وتحديات تطبيق تقنيات التعلم العميق واكتشاف الأجسام في تحليل السلوكيات المرتبطة بالتوحد.

Table of Contents:

Table of Figures :

Table of Shortcuts :

AI	Artificial inelegant
YOLO	You Only Look Once
SSBD	Self-Stimulatory Behaviours Dataset
mAP	Mean Average Precision

Table of Tables :

Chapter 1: Introduction

1.1 General Introduction

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition that affects communication, social interaction, and behavioral patterns in children. One of the major challenges in autism care is the early identification of behavioral signs, which plays a critical role in enabling timely intervention and improving long-term developmental outcomes.

With the rapid advancement of artificial intelligence, particularly in the field of computer vision, automated analysis of human behavior from video data has become increasingly feasible. Deep learning-based models have demonstrated strong capabilities in extracting meaningful patterns from visual inputs, offering new possibilities for behavior-based screening systems.

This project investigates the application of a real-time object detection model to identify autism-related behaviors from video data. The proposed system aims to support specialists by providing an automated tool that highlights potential behavioral patterns of interest.

1.2 Motivation

Early autism screening often depends on prolonged clinical observation and expert assessment, which may not always be accessible or scalable. In many cases, subtle behavioral cues can be overlooked, especially in non-controlled environments.

The motivation behind this project is to explore whether computer vision techniques can assist in detecting visible behavioral patterns associated with autism in a fast and non-intrusive manner. By leveraging real-time video analysis, the system aims to reduce reliance on manual observation and support early screening efforts.

1.3 Problem Definition

The primary problem addressed in this project is the lack of automated systems capable of detecting autism-related behaviors from video data in real time. Most existing approaches either require extensive datasets, specialized sensors, or operate in controlled environments.

This project focuses on designing a vision-based system that can detect predefined behavioral patterns directly from video frames, while operating under limited data availability and standard hardware constraints.

1.4 Objectives of the Project

The main objectives of this project are:

- To develop a video-based system for detecting autism-related behaviors.
- To investigate the suitability of YOLO-based object detection models for behavioral analysis.
- To evaluate the system using standard performance metrics such as Precision and Recall.
- To explore the feasibility of real-time deployment for supportive screening purposes.

1.5 Project Scope and Limitations

The scope of this project is limited to detecting visually observable behavioral patterns from video frames. The system does not aim to diagnose autism or replace professional clinical evaluation.
The proposed solution is intended as a supportive tool to assist specialists.
Limitations include the use of frame-based detection without explicit temporal modeling and reliance on a relatively small dataset.

1.6 Report Organization

This report is organized into several chapters. Chapter 2 presents background information and related work. Chapter 3 discusses autism spectrum disorder and behavioral characteristics. Chapters 4 to 7 describe the system design, dataset preparation, model training, and implementation details. Chapter 8 presents experimental results and discussion, followed by conclusions and future work in Chapter 9.

1.7 Tools and Technologies

Python was used as the main programming language due to its flexibility and strong support for machine learning and computer vision. YOLO v11 was employed as the object detection model for its real-time performance and suitability for video-based analysis. Roboflow was used for dataset management and annotation, while Google Colab provided a cloud-based environment with GPU support for training. OpenCV was utilized for video processing and frame extraction, and Matplotlib and Pandas were used for result visualization and analysis. Google Drive and GitHub were used to store, manage, and share the project files and trained models.

Chapter 2: Background and Related Work

2.1 Introduction

This chapter provides background information on autism spectrum disorder and reviews related research in autism behavior analysis. It discusses existing approaches and technologies, highlighting their limitations and the research gap addressed by this project.

2.2 Autism Spectrum Disorder Overview

Autism Spectrum Disorder is characterized by a wide range of behavioral manifestations, including repetitive movements, atypical gestures, and restricted patterns of activity. These behaviors vary significantly among individuals, making automated detection a challenging task.

Understanding these behavioral characteristics is essential for designing computer vision systems capable of capturing meaningful patterns from visual data.

2.3 Technology-Assisted Autism Detection

Several studies have explored the use of technology to support autism screening, including wearable sensors, eye-tracking systems, and video-based analysis. Among these approaches, video analysis offers a non-invasive and cost-effective solution capable of capturing natural behavior in real-world environments.

2.4 Computer Vision and Deep Learning

Computer vision enables machines to interpret visual information from images and videos. Deep learning models, particularly convolutional neural networks (CNNs), have shown strong performance in tasks such as object detection, activity recognition, and motion analysis.

2.5 Object Detection for Behavioral Analysis

Object detection models aim to localize and classify objects or regions of interest within an image. When applied to behavioral analysis, these models can be trained to detect specific body movements or gestures associated with particular behaviors.

2.6 YOLO-Based Detection Models

YOLO (You Only Look Once) is a single-stage object detection model designed for real-time performance. Its ability to process images in a single forward pass makes it suitable for applications requiring fast inference and immediate feedback.

2.7 Research Gap

Despite the increasing interest in applying computer vision techniques to autism-related behavior detection, one of the most significant challenges remains the availability and quality of suitable datasets. Most existing studies rely on relatively large, well-curated datasets collected under controlled conditions, which are often not publicly available or difficult to reproduce.
In practice, autism-related behavioral data is scarce, heterogeneous, and challenging to annotate, particularly when sourced from real-world videos.
Variations in recording conditions, camera angles, child behavior, and environmental context further complicate the data collection process.
This project addresses this research gap by investigating the feasibility of training a real-time behavior detection model under limited and imperfect data conditions. Significant effort was dedicated to manual annotation, data cleaning, and augmentation to compensate for data scarcity. By focusing on data preparation and robustness rather than ideal dataset assumptions, the project reflects realistic constraints encountered in practical applications.

Table 1 . Case Study 1

Title	Author	Journal	Year	Dataset Used	Training Approach	
Harnessing YOLOv11 for Enhanced Detection of Typical Autism Spectrum Disorder Behaviors Through Body Movements	Ayman Noor Hanan Almukhalf Arthur Souza Talal H. Noor	Diagnostics (MDPI)	2025	Custom dataset (72 videos – 13,640 images, 4 classes)	YOLOv1 1 Deep Learning Framewor k	**Results** Recall: 97% Precision: 96%, , F1-score: 97% Accuracy: 99%,

Table 2 . Case Study 2

Title	Author	Journal	Year	Dataset Used	Training Approach	Results
Advanced Gesture Recognition for Autism Spectrum Disorder Detection: Integrating YOLOv7, Video Augmentation, and VideoMAE for Naturalistic Video Analysis	Amit Kumar Singh Vrijendra Singh	arXiv	2024	SSBD – Self-Stimulator y Behavior Dataset	YOLOv7	Recall: 94% Precision: 93%, , F1-score: 94% Accuracy: 95%

Table 3 . Case Study 3

Title	Author	Journal	Year	Dataset Used	Training Approach	Results
Detection of Arm-Flapping Stereotypy in Children With Autism Using Mediapipe Landmarks and MobileNetV2 Features		Conference Paper CV4Health Workshop	2021	SSBD – Self-Stimulatory Behavior Dataset	MobileNetV2 Mediapipe Pose Landmarks LSTM	Poor Recall Poor Precision Accuracy: 60 %

Chapter 3: Autism Spectrum Disorder and Behavioral Characteristics

3.1 Introduction

This chapter introduces autism spectrum disorder and describes its key behavioral characteristics. It focuses on common movement patterns and behaviors relevant to autism detection, providing the conceptual foundation for the proposed system.

3.2 Autism Spectrum Disorder Overview

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterized by persistent challenges in social communication, restricted interests, and repetitive behaviors. The term “spectrum” reflects the wide variability in symptom severity and behavioral manifestations among individuals.

ASD is typically identified during early childhood; however, the manifestation of behavioral traits varies significantly depending on age, environment, and individual development. This variability makes early detection both critical and challenging.

3.3 Behavioral Characteristics Related to Autism

Children with autism often exhibit distinctive behavioral patterns that may include repetitive movements, atypical gestures, and unusual motor activities. These behaviors are not exclusive indicators of autism but may serve as important cues when observed persistently and in combination.

Examples of such behaviors include repetitive arm movements, body rocking, and other stereotypical motor patterns. These behaviors are visually observable and therefore suitable for video-based analysis.

3.4 Challenges in Behavioral Interpretation

One of the main challenges in autism-related behavior analysis is that similar movements may also appear in typically developing children. Context, frequency, and consistency play a significant role in distinguishing autism-related behaviors from normal activities.
As a result, automated systems must be designed with caution and should be considered supportive rather than diagnostic tools.

3.5 Visual-Based Behavioral Analysis

Video-based analysis provides a non-invasive method for observing children's behavior in natural environments. Unlike sensor-based approaches, video data captures both movement and context, allowing for more comprehensive behavioral interpretation.

This project focuses on extracting visual cues from video frames to identify specific behavioral patterns associated with autism.

3.6 Ethical Considerations in Behavioral Analysis

Working with autism-related data involves ethical considerations, including privacy, consent, and data sensitivity. For this reason, publicly available and ethically sourced data were used, and the system was designed to avoid personal identification.

Chapter 4: System Overview and Methodology

4.1 Introduction

This chapter presents an overview of the proposed system and explains the methodology followed in this project. It describes the overall workflow, from data preparation to model training and evaluation.

4.2 System Architecture Overview

The proposed system consists of a video input module, a behavior detection model, and an output visualization component. Video streams are processed frame by frame, where each frame is analyzed independently using a YOLO-based object detection model.

Detected behaviors are highlighted using bounding boxes, allowing visual interpretation of the model's predictions.

4.3 Choice of Detection Model

YOLO was selected due to its real-time performance and ability to detect multiple classes within a single frame. Unlike multi-stage detectors, YOLO processes each frame in a single forward pass, making it suitable for time-sensitive applications.

4.4 Frame-Based Video Processing

Although the input data consists of videos, the detection process operates at the frame level. Video streams are decomposed into individual frames, which are then processed independently by the model.
This approach simplifies the training process and allows compatibility with standard object detection architectures.

4.5 Real-Time Processing Considerations

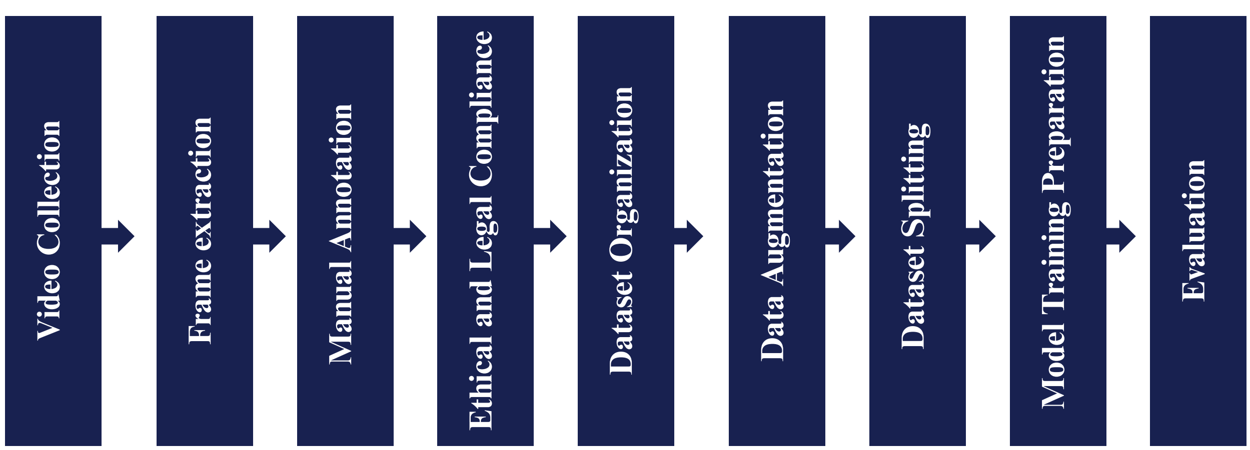
The system was designed with real-time deployment in mind. Model parameters such as input image size and batch size were selected to balance detection accuracy and inference speed.
This design choice enables the system to process video streams with minimal latency.

4.6 System Workflow

The system workflow includes the following steps:

1. Video Collection
2. Frame extraction
3. Manual Annotation
4. Ethical and Legal Compliance
5. Dataset Organization
6. Data Augmentation
7. Dataset Splitting
8. Model Training Preparation
9. Model Training and Evaluation

This structured workflow ensures modularity and scalability.



Chapter 5: Dataset Preparation and Annotation

5.1 Introduction

This chapter describes the dataset preparation process and the annotation methodology used in this project. It explains data collection, manual labeling, and preprocessing steps applied to prepare the dataset for model training.

5.2 Dataset Overview

The primary dataset used in this project is the Self-Stimulatory Behaviours in the Wild for Autism Diagnosis (SSBD) dataset. The dataset consists of 75 videos, each approximately 90 seconds long, depicting children exhibiting autism-related repetitive behaviors such as arm flapping, head banging, and spinning in natural, uncontrolled environments.

Due to the sensitive nature of autism-related data, the dataset is limited in size and exhibits significant variability in lighting conditions, backgrounds, camera angles, and subject movement.

5.3 Data Availability and Limitations

One of the main challenges encountered in this project was the limited availability of high-quality, annotated video data related to autism spectrum disorder. Publicly available datasets are scarce, and existing data often lack detailed annotations suitable for object detection tasks.
To address this limitation, additional data samples were carefully collected by the author from publicly accessible sources, strictly adhering to ethical guidelines, privacy regulations, and data usage policies. No private or identifiable personal information was used. These supplementary samples were integrated solely to enhance model learning and robustness.

5.4 Video Processing and Frame Extraction

Since the YOLO model operates on image-based inputs, all video data were converted into individual frames prior to training.

Frames containing relevant behavioral patterns were selectively extracted to ensure meaningful learning signals while reducing redundant or irrelevant data.

Each frame was resized and preprocessed to maintain consistency across the dataset while preserving essential visual features.

Figure : Samples of frames in DataSet

Figure 5.1

Figure 5.2

5.5 Manual Annotation Process

Manual annotation was performed to label autism-related behaviors within selected frames. Bounding boxes were drawn around regions of interest, and each instance was assigned a corresponding behavior class.

This process was conducted using Roboflow, which provided tools for precise annotation, class management, and dataset organization. Manual annotation was necessary due to the lack of pre-labeled datasets suitable for this task.

roboflow

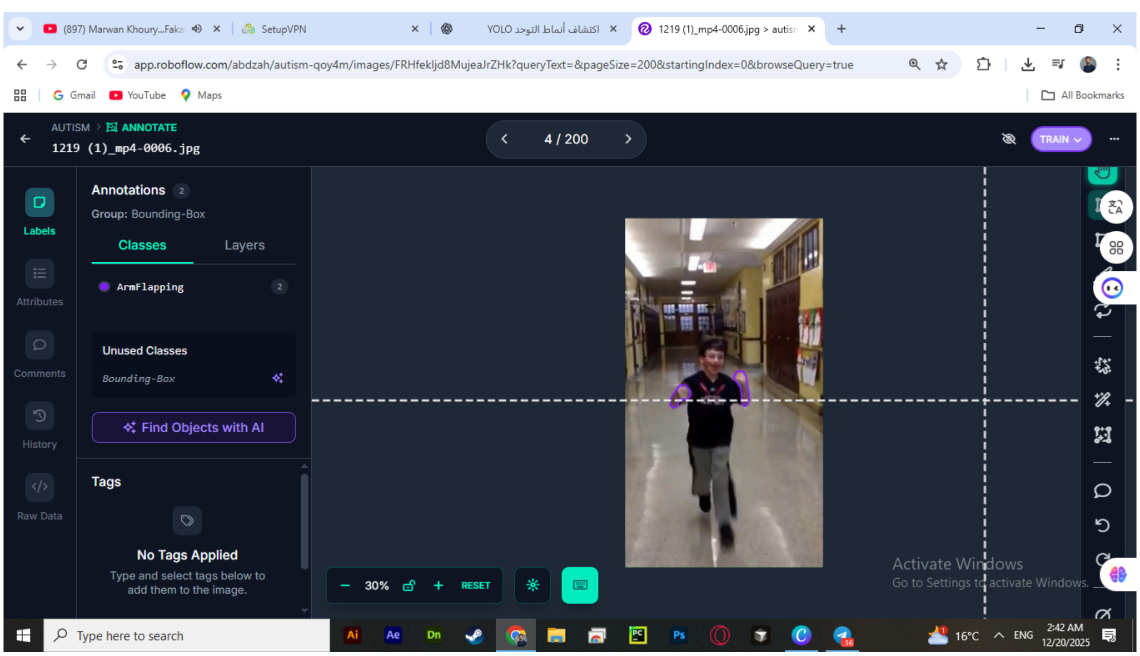


Figure 5.3: Manual Annotation in roboflow

5.6 Data Augmentation Techniques

Due to the limited size of the available dataset, data augmentation techniques were applied to improve model generalization and reduce overfitting. Several augmentation methods were used to increase data diversity while preserving the semantic meaning of autism-related behaviors. These techniques included horizontal flipping to account for directional movement variations, saturation and brightness adjustments to handle lighting differences, and slight blur to simulate motion and camera noise. The applied augmentations helped enhance the robustness of the model when dealing with real-world video data.

5.7 Dataset Splitting Strategy

Due to the very limited size of the available dataset, the data was divided only into training and validation sets, while a separate test set was not created. Allocating a test set under such data constraints could significantly reduce the amount of data available for learning and negatively affect model performance. Instead, the model was evaluated using the validation set during training, and additional external data samples collected independently were used for testing and qualitative evaluation. This strategy allowed better utilization of the limited labeled data while still providing a reliable assessment of the model's generalization ability.

5.8 Dataset Format and Organization

The processed dataset was exported in YOLO-compatible format, consisting of separate directories for images and labels, along with a configuration file defining class names and data paths.
This structure enables seamless integration with YOLO-based training frameworks.

Chapter 6: Model Training and Evaluation

6.1 Introduction

This chapter presents the training and evaluation process of the proposed autism behavior detection model based on the YOLO v11 architecture. It explains the training configuration, hyperparameter selection, dataset usage strategy, and evaluation methodology. Additionally, the chapter analyzes the obtained results and discusses the model performance under limited data conditions.

6.2 Model Selection

YOLO v11 was selected as the core detection model for this project due to its improved detection accuracy, efficient architecture, and suitability for real-time visual analysis. As a single-stage object detection framework, YOLO v11 performs object localization and classification simultaneously, making it highly effective for detecting autism-related behavioral patterns from video frames. Its fast inference speed and optimized design make it appropriate for practical and real-time applications.

6.3 Training Environment

The training process was conducted using Google Colab, leveraging GPU acceleration to improve training efficiency. This cloud-based environment enabled model training despite limited local computational resources. The annotated dataset, prepared and managed using Roboflow, was imported through a configuration file (data.yaml) compatible with the YOLO v11 training pipeline.

6.4 Training Configuration

The model was trained using an input image size of 640 × 640 pixels, which provides a balance between detection accuracy and computational efficiency. A batch size of 8 was selected to ensure stable training while fitting within GPU memory constraints. The model was trained for 25 epochs, a value chosen to reduce the risk of overfitting given the relatively small dataset size.

To further control overfitting, an early stopping mechanism was applied using a patience value of 5 epochs, allowing the training process to terminate automatically if validation performance stopped improving. Additionally, a cosine learning rate scheduler (cos_lr) was employed to gradually adjust the learning rate during training, promoting smoother convergence and improved generalization.

All training outputs, including model weights and performance logs, were saved to a specified project directory for reproducibility and future evaluation.

6.5 Dataset Usage and Splitting Strategy

Due to the limited size of the annotated dataset, the data was divided only into training and validation subsets. Creating a separate test set under these constraints could significantly reduce the amount of data available for learning. The validation set was used to monitor model performance during training, while external unseen data collected independently was used for post-training testing and qualitative evaluation. This strategy ensured efficient use of the available data while still allowing assessment of the model's generalization ability.

6.6 Evaluation Metrics

The model performance was evaluated using standard object detection metrics, including Precision, Recall, and mean Average Precision (mAP). Among these metrics, Recall was emphasized as the most critical measure in this project. In autism behavior detection, failing to detect relevant behaviors (false negatives) may reduce the practical usefulness of the system. Therefore, achieving a higher recall was prioritized to ensure that most autism-related behaviors were successfully identified.

6.7 Training Results

During training, both training and validation losses were monitored to evaluate convergence and detect signs of overfitting. The loss curves demonstrated a gradual reduction, indicating that the model effectively learned discriminative features from the dataset. The validation recall improved across training epochs, reflecting enhanced detection capability.

The final trained YOLO v11 model achieved a validation recall of approximately (0.7 - 0.86) , which represents a reasonable performance given the limited dataset size and the complexity of behavioral patterns. These results highlight the feasibility of using YOLO v11 for autism-related behavior detection under constrained data conditions.

6.8 Performance Analysis

The obtained results indicate that YOLO v11 can effectively detect autism-related behaviors from visual data. However, performance remains influenced by dataset size, behavioral variability, and the frame-based nature of the detection process. Some behaviors involve subtle or temporal motion patterns that are challenging to capture from individual frames.

Despite these limitations, the model demonstrates promising performance as a supportive analytical tool rather than a standalone diagnostic system. The results confirm the suitability of YOLO v11 as a foundation for future improvements and extended research.

6.9 Summary

This chapter presented the training and evaluation process of the YOLO v11-based autism behavior detection model. It discussed the training environment, configuration parameters, dataset usage strategy, evaluation metrics, and performance analysis. The findings demonstrate that YOLO v11 can be effectively applied to autism behavior detection even with limited data, while emphasizing the importance of data expansion and temporal modeling for future enhancements.

Chapter 7: Results and Discussion

7.1 Introduction

This chapter presents the experimental results obtained from training and evaluating the proposed YOLO v11-based autism behavior detection system. Both quantitative and qualitative results are analyzed to assess the effectiveness of the model. The chapter also discusses the limitations of the obtained results and evaluates the practical applicability of the system in real-world scenarios.

7.2 Quantitative Results

The performance of the trained YOLO v11 model was evaluated using standard object detection metrics, including Precision, Recall, and mean Average Precision (mAP). Among these metrics, Recall was considered the most critical indicator due to the nature of the problem.

The model achieved a validation recall of approximately 0.7, indicating that the system was able to correctly detect a significant portion of autism-related behavioral patterns present in the data. Precision and mAP values were relatively lower, which reflects the challenges associated with limited and highly variable behavioral data.

Although the numerical results do not represent state-of-the-art performance, they demonstrate that the model successfully learned meaningful visual patterns related to autism behaviors despite the constrained dataset size.

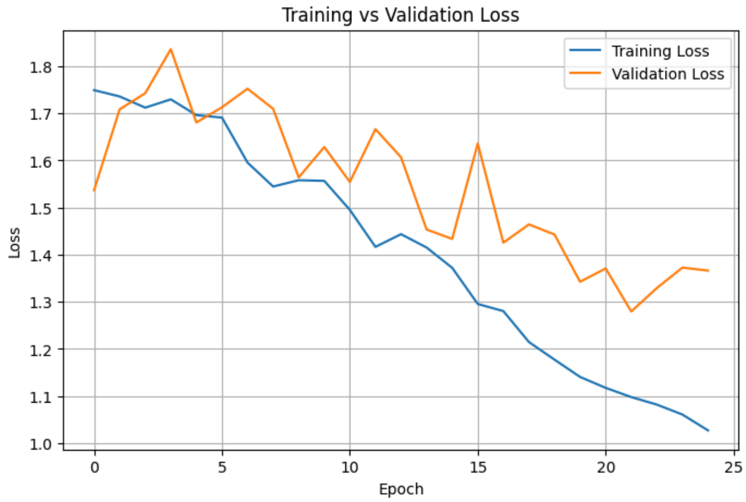


Figure 7.1: Training and validation loss curves of the YOLO v11 model

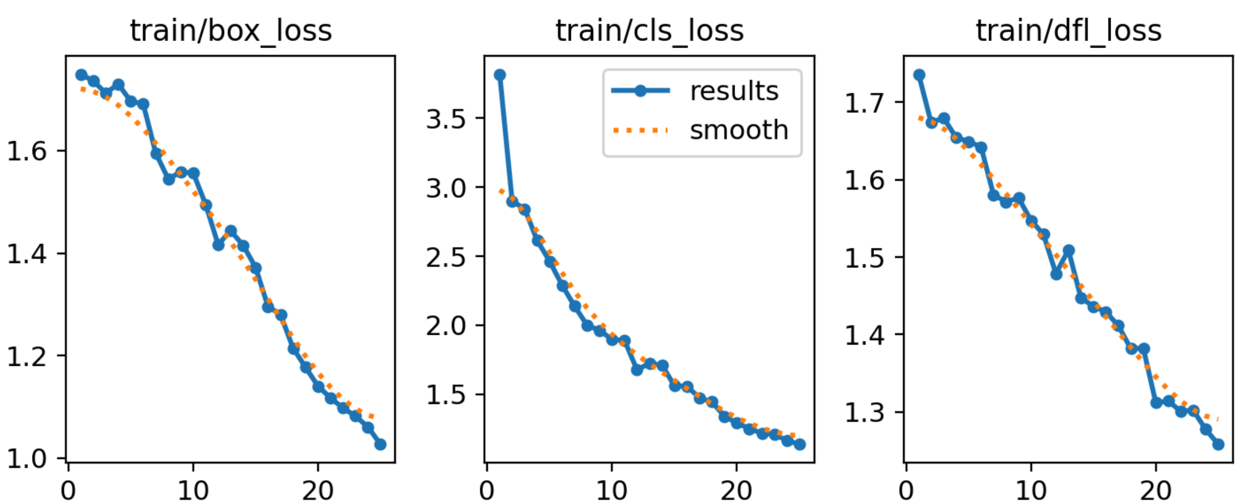


Figure 7.2: Training loss curves of the YOLO v11 model

7.3 Qualitative Results

In addition to numerical evaluation, qualitative analysis was conducted by visually inspecting detection outputs on sample images and video frames. The model was able to generate bounding boxes around several targeted behaviors, such as repetitive arm movements and abnormal motion patterns.

The qualitative results show that the model performs well in clear and well-lit scenarios, where movements are more distinguishable. However, detection accuracy decreases in cases involving subtle motions, partial occlusions, or low-quality video frames. These observations are consistent with the expected limitations of frame-based object detection models.

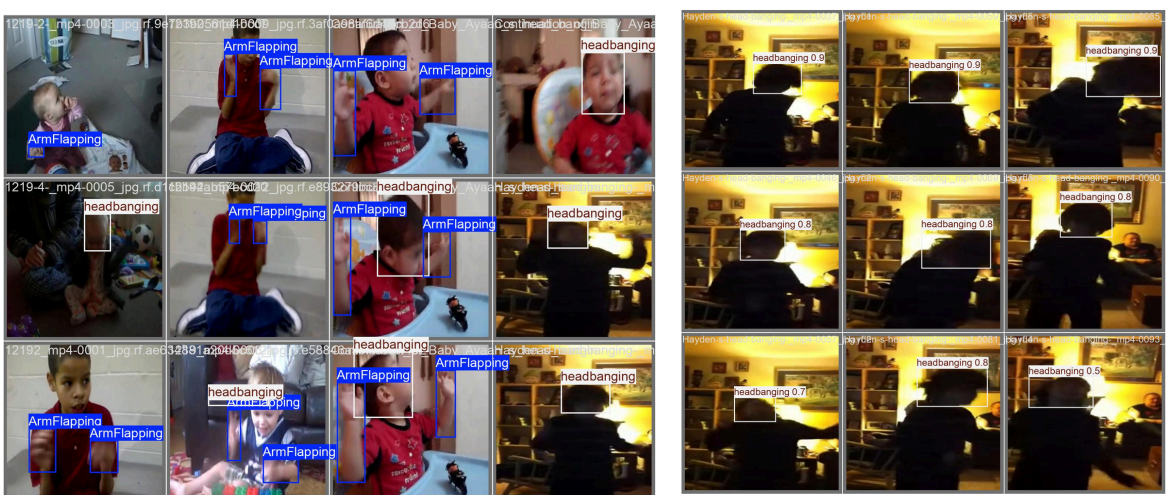


Figure 7.3: Sample detection results showing bounding boxes around autism-related behaviors

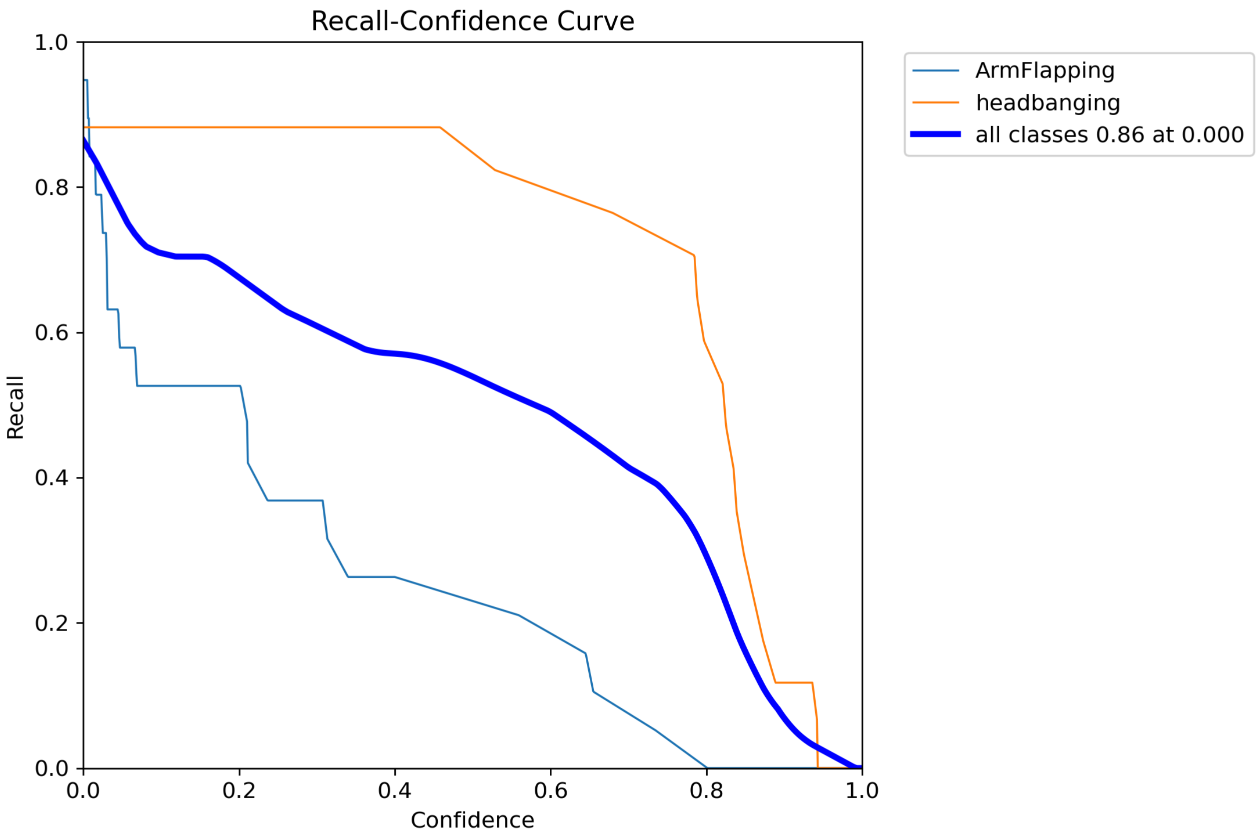


Figure 7.4: Recall - Confidence Curve

7.4 Discussion of Results

The obtained results indicate that YOLO v11 is capable of detecting autism-related behaviors from visual data; however, the performance is strongly influenced by data-related factors. The relatively moderate recall value can be primarily attributed to the limited size of the annotated dataset and the high variability of behavioral patterns among children.

Autism-related behaviors are not always visually distinct and may vary significantly between individuals. Moreover, many behaviors are defined by temporal repetition rather than single-frame appearance. Since the current system operates on individual frames without temporal modeling, certain behaviors may not be consistently detected.
Despite these challenges, the model demonstrates promising performance as a supportive tool for behavior analysis. The results confirm that real-time object detection models can serve as a foundation for automated autism behavior analysis when combined with sufficient data and advanced temporal modeling techniques.

7.5 Limitations of the System

Several limitations were identified during the experimental phase. The most significant limitation is the scarcity and quality of labeled data. Due to ethical and privacy constraints, collecting large-scale autism-related video data is particularly challenging.
Additionally, the system relies on manual annotations, which may introduce labeling inconsistencies. The absence of a dedicated test set and the reliance on external data for evaluation may also affect the reliability of performance assessment. Finally, the lack of temporal context limits the model's ability to capture behaviors that unfold over time.

7.6 Practical Implications

The proposed system is not intended to serve as a diagnostic tool. Instead, it should be viewed as a decision-support system that can assist specialists by highlighting potential behavioral patterns for further analysis. When used responsibly, such a system could reduce manual observation effort and support early behavioral screening in controlled environments.

7.7 Summary

This chapter presented and analyzed the results of the YOLO v11-based autism behavior detection system. Both quantitative metrics and qualitative observations indicate that the model can detect relevant behaviors despite limited data availability. The discussion highlighted key challenges and limitations while emphasizing the system's potential as a supportive analytical tool. These findings provide a strong basis for future improvements and extended research.

Chapter 8: Conclusion and Future Work

8.1 Introduction

This chapter concludes the presented work by summarizing the main contributions of the project and discussing its overall outcomes. It also outlines potential directions for future work that could enhance the system's performance and practical applicability.

8.2 Conclusion

This project presented a computer vision–based system for detecting autism-related behavioral indicators in children using the YOLO v11 object detection framework. The proposed approach focused on analyzing visual data extracted from videos to identify movement-based behaviors commonly associated with autism.

Despite the challenges posed by limited and heterogeneous data, the system demonstrated the ability to learn meaningful visual patterns through manual annotation, data augmentation, and careful training configuration. The achieved recall indicates that the model can detect a significant portion of relevant behaviors, supporting the feasibility of using real-time object detection models in this domain.

The results confirm that deep learning–based detection systems can serve as supportive tools for behavioral analysis rather than diagnostic instruments. The project highlights the importance of data quality, ethical data collection, and appropriate evaluation strategies when addressing sensitive medical and behavioral applications.

8.3 Future Work

Future work will focus on expanding and enhancing the dataset by collecting custom data in collaboration with special needs and autism care centers in Syria, subject to ethical approvals and data privacy regulations. Acquiring real-world data from controlled environments will significantly improve dataset quality, diversity, and representativeness, thereby strengthening model robustness and generalization.

In addition to visual data, future developments will explore a multimodal learning approach by incorporating audio signals, such as vocalizations and repetitive sounds commonly produced by individuals with autism. Integrating audio-based features alongside visual cues can provide complementary information that may improve behavior detection accuracy, particularly for behaviors that are not visually distinctive.

Furthermore, multimodal fusion techniques can be investigated to combine video and audio modalities within a unified deep learning framework. This extension has the potential to enhance detection reliability and move the system closer to practical deployment as a comprehensive behavioral analysis support tool.

Chapter 9: Refrences

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